

Douglas Williams

Technical Product Owner · AI-Assisted Software Delivery

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PROFILE

Technical builder with **28+ years** in mission-critical network operations who now delivers working software with AI-assisted, spec-driven development. I set the vision and requirements, direct AI coding agents (Kiro) to build in Python, and stay with the project through testing and debugging until it works. At CPKC I drove a company-wide event-analysis platform ingesting **billions of messages a day**. My operations depth means I understand the systems these tools serve.

EXPERIENCE

Senior PTC Applications Engineer

2022 – Present · Remote

CPKC (Canadian Pacific Kansas City)

- Own the vision and delivery of internal tools for Positive Train Control operations — safety-critical, where the tool has to work.
- Built LocoGrip, a company-wide event-analysis platform ingesting billions of PTC messages/day — now the primary Event & Incident Analysis tool for 45 users across 5 teams, automating log diagnosis that once took up to an hour per event.
- Manage 300+ mission-critical Linux & Windows servers running messaging and dispatching between the Train Management Dispatching System (TMDS) and locomotives — underpinning rail safety and on-time delivery.
- Delivered Python database-integration tools connecting SQL and Oracle sources into custom apps and reporting.
- Work spec-driven with Kiro: define requirements and target result, direct the AI, then test and debug to a working tool.

ATS Senior Network Engineer

Aug 2020 – Aug 2022

T-Mobile

- Product Lead for MCO (Message Controller) handling SMS/MMS traffic across virtualized server infrastructure.
- Migrated Sprint traffic to a new Replica MCO environment, consolidating onto one network.
- Authored MOPs and trained teams on new products at rollout.

Network Translations Specialist / NTAC Engineer

Jul 1998 – Aug 2020

Sprint

- Built a virtual SMS/MMS messaging platform (Mavenir, Rackspace); led SMS migration to IMS Core via SIP.
- Internalized MMS processing (Syniverse → virtualized MMSC), eliminating managed-services costs.
- Saved Sprint \$1.2M over three years via vendor renegotiation; ran 24/7/365 support.

SELECTED PROJECT

Oil & Gas Well-Tracker Application — React · Supabase · Python (own business, J&T Cook Corporation)

Full-stack app for working-interest tracking across 58 wells (48 producing) and 6 operators — investment vs. revenue, payout status, operator-statement ingestion, bank reconciliation, and tax reporting. Owned vision and requirements, directed AI to build, debugged to a live production deployment. github.com/djwilli69/well-tracker-demo

SKILLS & EDUCATION

AI-Assisted Development: spec-driven development, requirements, directing AI agents (Kiro), iterative test & debug

Product Ownership: vision → delivery, verifying output against real-world requirements

Tech shipped (AI-directed): React, Supabase (Postgres), Python, Vercel, SQL/Oracle, Git · **Foundation:** Linux/CLI, log analysis

Systems: Linux & Windows servers (300+ mission-critical), virtual servers, IMS Core · **Education:** B.S. Business Management, Friends University